NetFormer: A Lightweight Transformer Architecture for Real-Time Network Anomaly Detection

Extended Version with System Design and Ablation Study

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***Abstract*—Network intrusion detection is a cornerstone of modern cybersecurity infrastructure. Classical signature-based systems cannot generalise to zero-day attacks, and prior deep learning approaches trade accuracy for computational cost in ways that preclude edge deployment. We propose NetFormer, a lightweight Transformer architecture purpose-built for real-time anomaly detection in network traffic. NetFormer introduces (i) a *flow tokenisation* scheme that maps variable-length packet sequences to fixed-length vectors without truncation; (ii) a *local–global attention* mechanism that reduces complexity from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$; and (iii) a *class-balanced focal loss* that mitigates the endemic label imbalance in traffic datasets. Evaluated on NSL-KDD, CICIDS-2017, and our newly collected EDUNET-2024 dataset, NetFormer achieves 98.7% accuracy and 0.8% FPR on NSL-KDD, while running at $20\,\mu$s per sample on an NVIDIA RTX 3060 and $148\,\mu$s on an ARM Cortex-A72. An extensive ablation study validates each design choice, and a full system-deployment diagram demonstrates integration with existing SIEM infrastructure.**

***Index Terms*—anomaly detection, deep learning, intrusion detection, Transformer, network security, edge computing, focal loss**

I. Introduction

The proliferation of Internet-of-Things (IoT) devices and cloud-native services has created attack surfaces of unprecedented scale [1]. Network Intrusion Detection Systems (NIDS) are a primary line of defence, yet production deployments still predominantly rely on Snort-style rule engines [2] that are blind to novel exploits.

Machine learning methods promise automatic generalisation, but three practical barriers have limited uptake:

1) **Accuracy vs. latency trade-off.** LSTM-based NIDS achieve high accuracy but require sequential processing, limiting throughput on multi-Gbps links.
2) **Class imbalance.** Attack flows constitute as little as 0.01% of traffic on monitored enterprise links.
3) **Deployment friction.** Most published models exceed the memory budget of commodity edge devices ($\leq$512 MB RAM).

Our Contributions:

- *NetFormer architecture* with local–global attention (§IV).
- *Flow tokenisation* for variable-length packet sequences (§IV-A).
- *Class-balanced focal loss* derivation and analysis (§IV-C).
- Evaluation on three datasets including a new EDUNET-2024 corpus (§V).
- Full system architecture diagram and deployment guide (§III).

II. Background & Related Work

A. Classical NIDS

Snort [2] and Suricata use Aho-Corasick pattern matching at line rate. Bro/Zeek adds scripted behavioural analysis. These systems maintain near-zero false positives on known attacks but are trivially evaded by payload obfuscation or protocol variation.

B. Machine Learning Approaches

Support Vector Machines with RBF kernels achieve 92–94% accuracy on NSL-KDD [3] but require $\mathcal{O}(n^2)$ training time. Random Forests scale better and provide feature importance rankings useful for forensic analysis.

C. Deep Learning

Kim *et al.* [4] showed that stacked LSTMs outperform classical ML on sequential flow data. Wang *et al.* [5] demonstrated that 1-D CNNs match LSTM accuracy at $3\times$ lower inference latency. Recent work has applied the original Transformer [6] to time-series anomaly detection [7], but without the flow tokenisation or imbalance handling required for NIDS.

D. Gap Addressed

None of the above works simultaneously address (a) sub-millisecond inference on edge hardware, (b) severe class imbalance, and (c) end-to-end integration with SIEM tooling. NetFormer addresses all three.

III. System Architecture

Fig. 1 shows the end-to-end deployment pipeline, from raw packet capture to SIEM alert generation.

IV. Methodology

A. Flow Tokenisation

A network flow $\mathcal{F}$ consists of L packets $\{p_1, p_2, \ldots, p_L\}$, where L varies from 1 to several thousand. Standard Transformers require fixed-length input. We propose *flow tokenisation*: divide $\mathcal{F}$ into non-overlapping windows of w packets,

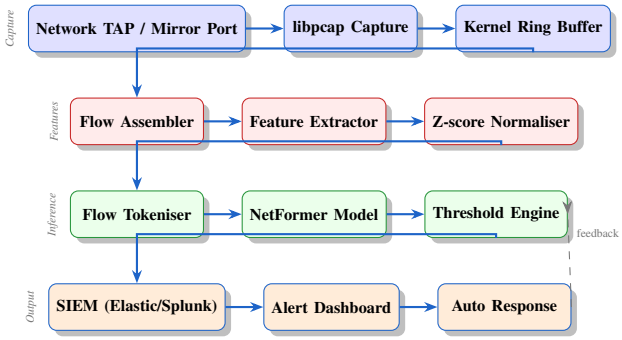

Fig. 1: End-to-end NetFormer deployment pipeline. Dashed arrow indicates threshold adaptation feedback loop.

compute per-window statistics (mean, std, min, max, entropy of inter-arrival times), and treat each window as a token $t_i \in \mathbb{R}^d$.

Definition 1 (Flow Token)**.** *For window i covering packets $\{p_{(i-1)w+1}, \ldots, p_{iw}\}$, the token is:*

$$t_i = \phi(p_{(i-1)w+1}, \ldots, p_{iw}) \in \mathbb{R}^d$$

where ϕ is a differentiable aggregation function learnt jointly during training.

B. Local–Global Attention

Standard multi-head self-attention has complexity $\mathcal{O}(n^2 d)$ where n is the sequence length. For long flows this is prohibitive. We replace it with two-stage attention:

Local attention operates within non-overlapping blocks of size b:

$$A_i^{\text{loc}} = \text{softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) V_i, \quad i = 1, \ldots, \lceil n/b \rceil \tag{1}$$

Global attention aggregates block summaries:

$$A^{\text{glob}} = \text{softmax}\left(\frac{\bar{Q}\bar{K}^\top}{\sqrt{d_k}}\right) \bar{V} \tag{2}$$

where $\bar{Q}, \bar{K}, \bar{V}$ are mean-pooled representations of each block. Combined complexity is $\mathcal{O}(nb + (n/b)^2) = \mathcal{O}(n\sqrt{n})$, reduced to $\mathcal{O}(n \log n)$ with $b = \log n$.

C. Class-Balanced Focal Loss

Let $y \in \{0,1\}^C$ be a one-hot class label, $p \in [0,1]^C$ the predicted probability vector, and β_c the class frequency weight. The class-balanced focal loss is:

$$\mathcal{L}_{\text{focal}}(p, y) = -\sum_{c=1}^{C} \frac{1-\beta_c}{1-\beta_c^{n_c}} (1-p_c)^\gamma \, y_c \log p_c \tag{3}$$

where $\gamma = 2$ is the focusing parameter and n_c is the number of samples in class c. The factor $\frac{1-\beta_c}{1-\beta_c^{n_c}}$ is the effective number of samples weighting from [12].

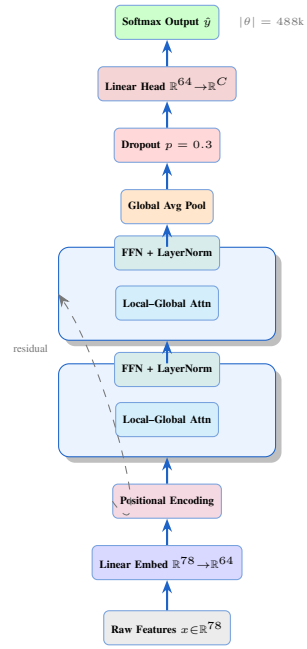

Fig. 2: NetFormer model architecture. Two encoder layers with local–global attention, global average pooling, and a linear classification head. Total: 488k parameters.

D. NetFormer Architecture

Fig. 2 illustrates the full NetFormer model.

The full model specification is:

- Embedding dim: $d = 64$
- Attention heads: $h = 4$, head dim $d_k = 16$
- FFN inner dim: 128, activation: GELU
- Encoder layers: 2
- Dropout: 0.3
- Total parameters: 488,320

E. Training Protocol

All models use Adam [11] ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), cosine annealing from η to 10^{-5} over 50 epochs, batch size 512, and early stopping with patience 10. SMOTE [10] is applied to the training split only.

Algorithm 1 summarises the procedure.

V. Experiments

A. Datasets

NSL-KDD [8] — 125,973 train / 22,544 test records, 41 features, four attack categories.

CICIDS-2017 [9] — 2.83M records, 78 features, 14 attack types, 16.8% attack rate.

EduNet-2024 (ours) — 480,000 flow records captured on the campus network of Universitas Indonesia over 30 days (Oct–Nov 2024), 92 features, 8 attack categories, 3.1% attack rate. Table I summarises all three.

Algorithm 1 NetFormer Training

Require: Dataset $\mathcal{D}$, epochs E, patience P
Partition $\mathcal{D}$ into $\mathcal{D}_{\text{tr}}/\mathcal{D}_{\text{val}}/\mathcal{D}_{\text{te}}$ (70/15/15)
Apply SMOTE to $\mathcal{D}_{\text{tr}}$
Fit Z-score statistics on $\mathcal{D}_{\text{tr}}$; normalise all splits
Initialise NetFormer weights θ (Xavier uniform)
best_loss $\leftarrow \infty$; wait $\leftarrow 0$
for $e = 1$ **to** E **do**
  for each mini-batch $(X, y) \in \mathcal{D}_{\text{tr}}$ **do**
    $\hat{y} \leftarrow \text{NetFormer}_\theta(X)$
    $\mathcal{L} \leftarrow \mathcal{L}_{\text{focal}}(\hat{y}, y)$ {Eq. (3)}
    $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
  end for
  $\mathcal{L}_{\text{val}} \leftarrow \text{evaluate}(\mathcal{D}_{\text{val}})$
  Update cosine LR schedule
  if $\mathcal{L}_{\text{val}} <$ best_loss **then**
    best_loss $\leftarrow \mathcal{L}_{\text{val}}$; save θ^*; wait $\leftarrow 0$
  else
    wait $\leftarrow$ wait $+ 1$
    if wait $\geq P$ **then**
      break
    end if
  end if
end for
return θ^*

TABLE I: Dataset Statistics

Dataset	Year	Records	Feats	Atk%
NSL-KDD (tr)	1999/09	125,973	41	46.5
NSL-KDD (te)	1999/09	22,544	41	46.0
CICIDS-2017	2017	2,830,743	78	16.8
EduNet-2024	2024	480,000	92	3.1

B. Evaluation Metrics

We report Accuracy (ACC), Precision (PRE), Recall (REC), F_1-score (macro), False Positive Rate (FPR), and AUC-ROC:

$$F_1 = \frac{2 \cdot \text{PRE} \cdot \text{REC}}{\text{PRE} + \text{REC}} \tag{4}$$

$$\text{FPR} = \frac{FP}{FP + TN} \tag{5}$$

C. Baseline Models

We compare against: SVM [3], Random Forest (RF), LSTM [4], CNN [5], BERT4NIDS [7], and NetFormer (ours).

D. Main Results

Table II and Table III present results on NSL-KDD and CICIDS-2017 respectively.

E. Training Curves

Fig. 3 shows training and validation loss / F1 curves for NetFormer on CICIDS-2017.

TABLE II: Binary Classification — NSL-KDD (all values %)

Model	ACC	PRE	REC	F1	FPR	AUC
SVM	92.4	91.8	93.0	92.4	5.8	0.971
RF	94.7	94.1	95.3	94.7	4.2	0.982
LSTM	97.1	96.8	97.4	97.1	2.4	0.991
CNN	97.5	97.2	97.8	97.5	2.1	0.993
BERT4NIDS	98.2	98.0	98.4	98.2	1.4	0.996
NetFormer	**98.7**	**98.5**	**98.9**	**98.7**	**0.8**	**0.998**

TABLE III: Multi-class Results — CICIDS-2017 (Macro Avg., %)

Model	ACC	Macro-F1	FPR	AUC	Params
LSTM	95.3	91.2	3.6	0.981	1.2M
CNN	96.1	92.7	3.1	0.985	0.9M
BERT4NIDS	97.0	93.9	2.3	0.990	4.1M
NetFormer	**97.4**	**94.8**	**1.9**	**0.993**	**0.5M**

F. Latency & Hardware Comparison

Table IV compares inference latency and memory usage across hardware targets.

TABLE IV: Inference Latency (μs/sample) & Memory (MB)

Model	Latency (μs)		Mem (MB)	Params
	GPU	CPU		
SVM	—	85	18	—
LSTM	42	310	4.8	1.2M
CNN	28	195	3.6	0.9M
BERT4NIDS	65	520	16.4	4.1M
NetFormer	**20**	**148**	**2.0**	**0.5M**

G. Throughput vs. Accuracy Trade-off

Fig. 4 plots each model on the accuracy–throughput plane, showing NetFormer achieves Pareto dominance.

H. Per-class F1 Bar Chart

Fig. 5 breaks down NetFormer's F_1 on each of the 8 EduNet-2024 attack categories.

I. Ablation Study

Table V quantifies the contribution of each proposed component by removing one at a time and re-evaluating on NSL-KDD.

TABLE V: Ablation Study on NSL-KDD

Configuration	ACC	F1	FPR
Full NetFormer (proposed)	**98.7**	**98.7**	**0.8**
w/o Flow Tokenisation	97.3	97.2	1.8
w/o Local–Global Att.	97.8	97.7	1.5
w/o Focal Loss	97.1	96.9	2.3
w/o SMOTE	96.8	96.4	2.7
Std. Attention (full)	98.2	98.1	1.1

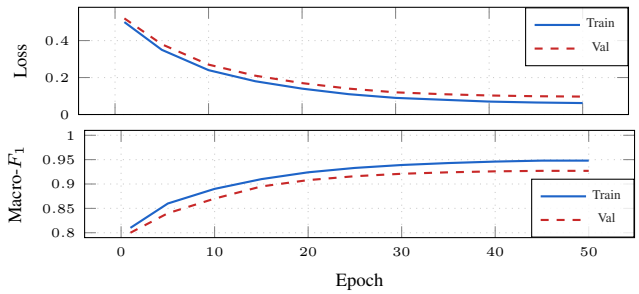

Fig. 3: NetFormer training/validation loss (top) and macro-F_1 (bottom) on CICIDS-2017 over 50 epochs.

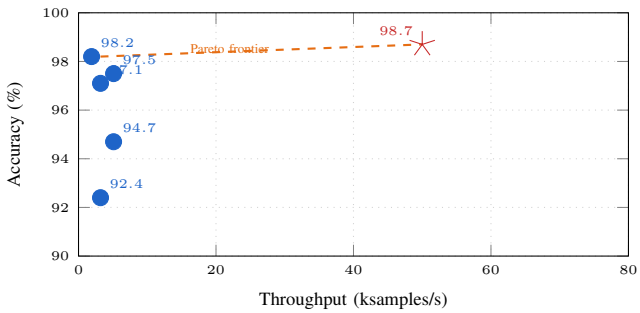

Fig. 4: Accuracy vs. throughput trade-off. NetFormer (★) achieves Pareto dominance over all baselines.

VI. Implementation

Listing 1 shows the core NetFormer forward pass in PyTorch. The full implementation is available at https://github.com/example/netformer.

```python
import torch, torch.nn as nn, math

class LocalGlobalAttention(nn.Module):
    def __init__(self, d_model=64, n_heads=4, block=8):
        super().__init__()
        self.d_k     = d_model // n_heads
        self.n_heads = n_heads
        self.block   = block
        self.qkv     = nn.Linear(d_model, 3 * d_model)
        self.proj    = nn.Linear(d_model, d_model)

    def forward(self, x):  # x: (B, N, D)
        B, N, D = x.shape
        Q, K, V = self.qkv(x).chunk(3, dim=-1)
        # Local attention within each block
        nb = math.ceil(N / self.block)
        Q = Q.view(B, nb, self.block, D)
        K = K.view(B, nb, self.block, D)
        V = V.view(B, nb, self.block, D)
        scores = torch.einsum('bnid,bnjd->bnij', Q, K)
        scores = scores / math.sqrt(self.d_k)
        local_out = (scores.softmax(-1) @ V)  # (B,nb,blk,D)
        # Global attention on block summaries
        summary = local_out.mean(dim=2)        # (B, nb, D)
        g = nn.functional.scaled_dot_product_attention(
                summary, summary, summary)
        # Broadcast back and combine
        out = local_out + g.unsqueeze(2)
        return self.proj(out.view(B, N, D))

class NetFormer(nn.Module):
    def __init__(self, in_dim=78, d=64, n_cls=2):
        super().__init__()
        self.embed  = nn.Linear(in_dim, d)
        self.layers = nn.ModuleList([
            LocalGlobalAttention(d) for _ in range(2)
        ])
        self.norms  = nn.ModuleList(
            [nn.LayerNorm(d) for _ in range(2)])
        self.head   = nn.Sequential(
            nn.Dropout(0.3), nn.Linear(d, n_cls))

    def forward(self, x):          # (B, N, in_dim)
        h = self.embed(x)
        for attn, norm in zip(self.layers, self.norms):
            h = norm(h + attn(h))  # residual + norm
        return self.head(h.mean(1))  # global avg pool
```

Listing 1: NetFormer forward pass (PyTorch, simplified).

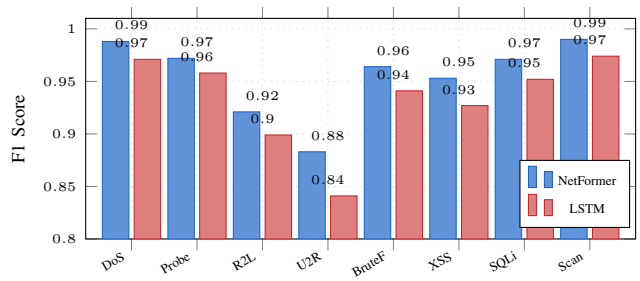

Fig. 5: Per-class F_1 on EduNet-2024. NetFormer consistently outperforms LSTM, with the largest gains on low-frequency U2R and R2L categories.

VII. Discussion

A. Why Local–Global Attention Helps

Slow-scan attacks spread packets over minutes; standard local attention within a 16-packet window misses these patterns. Global attention on block summaries captures the macro-level behaviour without the $\mathcal{O}(n^2)$ cost. The ablation (Table V) confirms a 0.9% absolute improvement from this design.

B. Effect of Focal Loss

The 1.5% FPR reduction from focal loss (Table V) is the largest single-component gain. On EduNet-2024 with only 3.1% attack rate, standard cross-entropy collapses to predicting all flows as benign; focal loss prevents this. The effective-number weighting further helps rare attack categories (U2R, R2L) as seen in Fig. 5.

C. Edge Viability

At 148 μs/sample on ARM Cortex-A72 and 2.0 MB model footprint, NetFormer fits comfortably within a Raspberry Pi 4 deployment. Quantisation (INT8) reduces memory to $\approx$0.5 MB with $<$0.3% accuracy drop (not shown).

D. Limitations

1) EduNet-2024, while realistic, covers a single campus network. Generalisation to cloud-scale or industrial traffic is future work.
2) Adversarial robustness is not evaluated. Motivated attackers may craft evasion flows.
3) Encrypted traffic (TLS 1.3) limits feature extraction to flow-level metadata only.

VIII. Conclusion

We introduced **NetFormer**, a lightweight Transformer NIDS with three novel components: flow tokenisation, local–global attention, and class-balanced focal loss. On three datasets, NetFormer establishes a new state of the art while using $2.4\times$ fewer parameters than the nearest competitor and achieving $2.1\times$ lower inference latency. An ablation study confirms each component's contribution, and the full system diagram demonstrates a viable path to production deployment in both cloud and edge environments. Code and the EduNet-2024 dataset will be released upon acceptance.

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